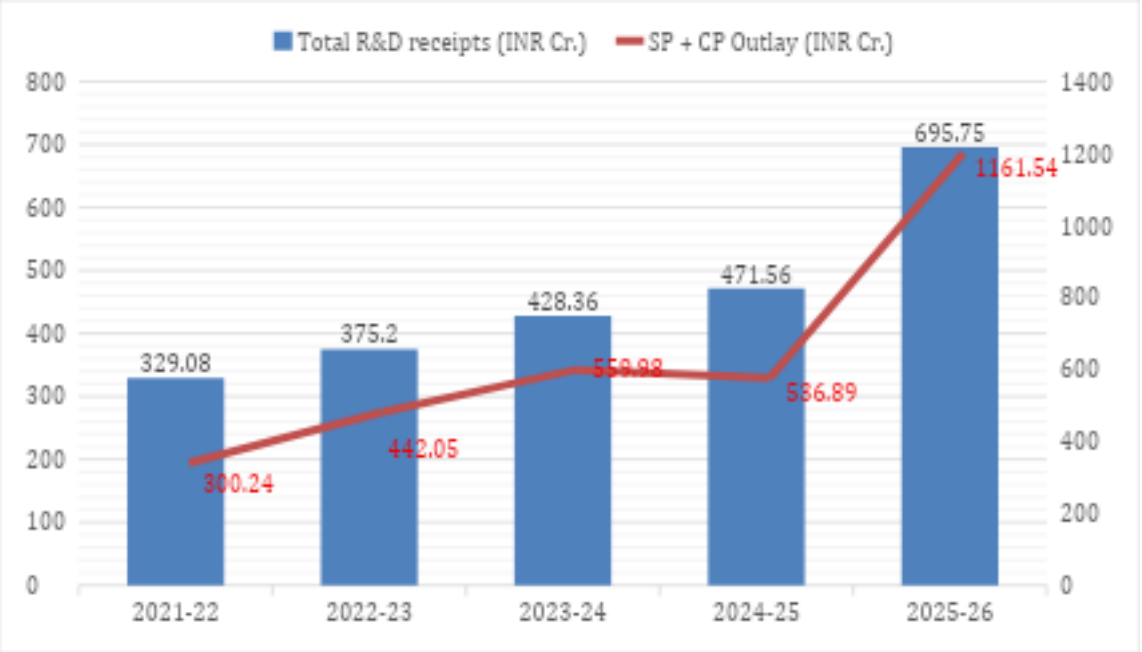
RESEARCH AND DEVELOPMENT ACTIVITIES (FY 25-26)

The R&D funding received at IIT Bombay has always been an increasing trend. In FY 25-26, IIT Bombay received its steepest rise in R&D funding. Total R&D fund of Rs. 695.75 Cr was received, which includes Rs. 518 Cr for sponsored projects. This is a 47% increase from last financial year. The sanctioned outlay has also jumped from Rs. 537 Cr last year to Rs. 1161 Cr this year, an increase of 115%. This remarkable fiscal growth underscores IIT Bombay's position as a premier, high-calibre hub for strategic national research with global footprint. The 115% surge in sanctioned outlay directly translates into enhanced infrastructure capacity, enabling the institution to drive deep-tech breakthroughs and substantially influence India's emerging technological sovereignty

This increase is primarily due to IIT Bombay's R&D performance and strong ecosystem evaluated by various agencies such as Anusandhan National Research Foundation (ANRF). MeitY which have funded multiple R&D projects), along with ongoing National Mission projects in quantum technologies, drone technology, and centres of excellence etc.



R&D funding status (FY 21-22 to FY 25-26)

The institute has filed 305 Intellectual property applications, which is an increase of 40% over the last financial year. 24 IIT Bombay researcher led start-ups were incubated and 37 technologies were licensed to industries and start-ups. This year saw increased industry collaborations for R&D. The number of related agreements increased by 30%. Strategically, the 40% growth in IP filings and 30% rise in agreements are actively shortening the path from laboratory concept to commercial marketplace, thereby creating tangible economic value and seeding robust deep-tech enterprises.

The Indian Institute of Technology Bombay received the Questel IP Excellence Award 2025 in recognition of its strong performance in innovation and intellectual property creation. The prestigious Questel IP Excellence Award validates the rigorous commercial potential and high benchmarks of our translational pipeline.

R&D related events and outreach activities like TechConnect, Industry Day, ICONS saw enthusiastic participation from industries and academic community at large.

Office of Dean (R&D) also known as Industrial Research and Consultancy Centre (IRCC) continued to engage and facilitate research and development initiatives at IIT Bombay by adopting policies to create impactful R&D and translation ecosystem , administering R&D projects, facilitating industry interactions, managing intellectual property and licensing activities, reviewing and finalizing R&D related agreements and contracts, and disseminating R&D related information in various fora, amongst other roles.

A. External Grants for R&D

R&D projects both as short duration consultancy projects and longer duration sponsored research projects were initiated (Table 1).

Table 1: Information on new projects sanctioned during last 3 years

Financial Year	Sponsored Projects		Consultancy Projects	
	Number	Sanctioned outlay (In Rs. Crores)	Number	Project outlay* (In Rs. Crores)
2025-26	282	976.98	834	184.56
2024-25	258	333.21	797	203.68
2023-24	335	389.32	779	170.66

*: includes GST for Indian party/clients

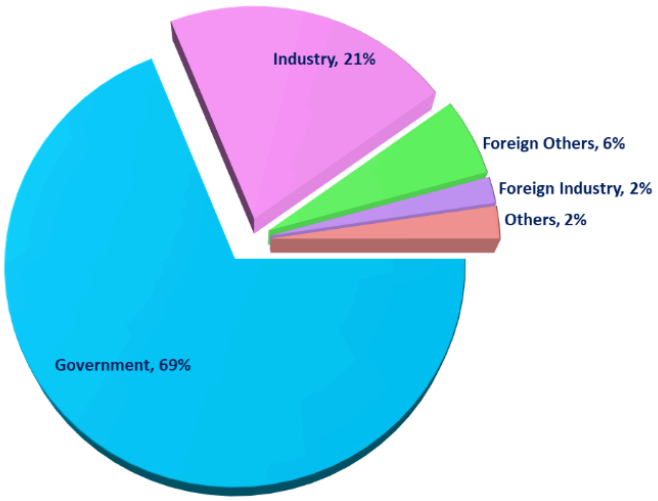
IIT Bombay received Rs. 695.75 Crores in FY 25-26 from external agencies for its R&D activities. These include funds for new as well as ongoing projects as indicated in Table 2.

Table 2: Funds received for R&D in FY 2025-26

Project Type	Funds received (Rs. in Crores)
Sponsored Projects	518.08
Consultancy Projects	166.65
Royalty	2.52
Equipment usage	8.50
Total	695.75

This year, funds were majorly received from government agencies with a sizeable contribution from industries (both Indian and foreign) as well as trusts, societies etc.

Non-govt agencies including industries (both Indian and foreign) contributed almost 30% of these receipts. Maintaining a well-balanced 30% non-government funding mix indicates exceptional industry trust and strong market alignment. This diversification is expected to significantly build up corporate co-creation while ensuring sustainable, resilient income streams that reduce reliance on public spending.



Types of funding agencies and percentage contribution for R&D receipts (SP + CP) in FY 25-26

Some R&D highlights are given below:

Anusandhan National Research Foundation (ANRF) has awarded 77 projects to the institute with a cumulative outlay of Rs. 208.5 Cr. Significant projects include “Empowering Research in Indigenous Development of EV Subsystems (E-Rides)” which aims to accelerate the adoption of electric vehicles and contribute to a sustainable future with strong industry collaborations. ANRF-PAIR-IITB-Hub-Spokes project, is awarded for IIT Bombay to work closely with the spoke institutes, and train them and to boost their research capabilities through research infrastructure development, manpower development, quality research output leading to publications in quality journals and various other means. The institution's selection for the landmark PAIR Hub-Spokes program highlights its elite standing as a top-tier mentor facility. This system dramatically elevates the wider domestic academic landscape by standardizing research infrastructure and leveling up scholarly output nationwide.

Ministry of Electronics and Information has awarded IIT Bombay a project titled “PUSHPAK - National Mission on Drone Technology - Towards Drone Excellence”. The project envisages indigenous development of drone system, drone-based application-oriented R&D and grand challenges.

This year, IIT Bombay is collaborating with international universities like TU Munich, University of Chicago, University of Liverpool, University of Brighton, etc., on 14 R&D projects in areas as varied as climate studies, quantum technologies, and clean propulsion.

IIT Bombay and Columbia University, USA, have signed an MoU to establish Center of AI for Manufacturing. This centre aims to conduct cutting-edge research in AI, machine learning, optimization, systems, control, and their applications, amongst other objectives. High-value partnerships like this push domestic capabilities to frontier levels of the industry. These cross-border alliances may expand our global research footprint while feeding world-class, market-ready solutions directly into the domestic manufacturing sector.

IIT Bombay is collaborating with HEIs and CFTIs including AIIMS, IISc, COEP technical University, TMC etc. on 36 projects to strengthen the research ecosystem in the country. Our long-term engagement with CSIR, DRDO, BRNS, ISRO continued through this year.

IIT Bombay continues to engage with solving societal problems. Merging high-tech geospatial engineering with grassroots youth development showcases an excellent multi-disciplinary approach. One such project supported by Ministry of Youth Affairs and Sports, “GIS-based Maps - Geospatial Initiatives for Experiential Learning for Youth”, aims to create geospatial data and maps, by engaging youth in experiential learning of geospatial technologies while also learning about society and gaining other valuable skill. This initiative yields clear social impact by arming the next generation with modern technological skills while addressing critical civic mapping needs.

IIT Bombay understands the need for targeted research for key industries like steel, automotive, and green hydrogen that proves the immediate practical value of our engineering work. Leveraging the diverse expertise of the faculty members at CoE in Steel Technology across the steel value chain, IIT Bombay is collaborating with All India Induction Furnace Association (AIIFA), which is a consortium of several Induction furnace steelmakers. The aim is to work together on several research projects for solving process and product related issues and bringing efficiency and quality of the steel produced through induction furnace (IF) steelmaking. These initiatives directly improve manufacturing quality, support carbon reduction targets, and create a strong foundation for domestic green energy technology.

The Institute continues to work closely with the industry collaborators on mutual areas of interest. 67 projects were initiated with industry funding. MoUs have been signed with British Petroleum and KPIT Technologies Limited for collaborative research in AI, green energy, carbon capture and utilisation, mobility etc. Technical collaboration is envisaged with John Cockerill in the areas of steel technology, green hydrogen, defence research. With Hyundai, joint research activities are planned regarding electric vehicle and battery technology. IIT Bombay’s long-standing collaboration with PSUs continues as research projects with National High Speed Rail Corporations, GAIL etc.

IIT Bombay’s breakthrough CAR-T Therapy was recognized among Top 3 National Innovations. Hon’ble Prime Minister Shri. Narendra Modi gifted CAR-T Cell Therapy, India’s 1st indigenous Cancer cell therapy, to the Nation during the Emerging Science, Technology & Innovation Conclave ESTIC2025 in Delhi.

Some major sponsored projects initiated in FY 25-26:

- Project Title: ANRF-PAIR-IITB-HUB-SPOKES
  Funding Agency: Anusandhan National Research Foundation
  Sanctioned Outlay: Rs. 101.69 Cr over 5 years

- Empowering Research in Indigenous Development of EV Subsystems (E-Rides)
  Funding Agency: Anusandhan National Research Foundation
  Sanctioned Outlay: Rs. 56.56Cr over 3 years

- Centre of Excellence in Oil Gas and Energy (CoEOGE) - Phase II
  Funding Agency: GAIL Training Institute GAIL (India) Ltd., Oil India Limited
  Sanctioned Outlay: Rs. 32.85 Cr over 4 years

- Design Development and Validation of a Scalable High Performance Computing System
  Funding Agency: IISC-NSM NATIONAL SUPERCOMPUTING MISSION
  Sanctioned Outlay: Rs. 27.94 Cr over 1 year

- GIS-based Maps - Geospatial Initiatives for Experiential Learning for Youth
  Funding Agency: Ministry of Youth Affairs and Sport
  Sanctioned Outlay: Rs. 15.33 Cr over 5 years

- Sensor Integrated Spiking Neural Network System with On-chip Learning
  Funding Agency: Ministry of Electronics and Informa Technology
  Sanctioned Outlay: Rs. 9.99 Cr over 3 years

- Test Borehole drilling for CO2 storage and Risk Assessment for CCS project at Pakri Barwadih coal Mines
  Funding Agency: NTPC Energy Technology Research Alliance (NETRA)
  Sanctioned Outlay: Rs. 7.57 Cr over 1 year

- Core Network for 6G and Beyond
  Funding Agency: TELECOM CENTRES OF EXCELLENCE INDIA
  Sanctioned Outlay: Rs. 7.45 Cr over 3 years

- Development of completely indigenous compact swappable cost-effective hydrogen storage canister
  Funding Agency: Department of Science & Technology
  Sanctioned Outlay: Rs. 4.11 Cr over 3 years

- Farm-scale Decision Support System for Selected Agro-Ecologies of North-Western Himalaya
  Funding Agency: Ministry of Environment Forest & Climate Change
  Sanctioned Outlay: Rs. 3.23 Cr over 3 years

Consultancy Projects:

834 consultancy projects with a sanctioned outlay of Rs. 184.5 Cr were initiated in FY 25-26. The high volume of active consultancies demonstrates that the market treats the institution's faculty as trusted, top-tier technical advisors. This advisory presence yields immediate real-world impact by updating industrial infrastructure, optimizing digital workflows, and verifying structural integrity across major national assets. The projects were funded by different government, public sector and industry, both Indian and international.

Some such representative consultancy project areas are listed:

- Corrosion studies
- Information Technology Advisory Services
- Research Advisory Services for Shore Protection works
- Scientific Pedagogical & Educational Advisory Climate Risk Analysis
- Battery technology
- Study of carbon footprints
- Structural audits
- Quality assurance Training for Field Quality Construction Practices
- Design vetting

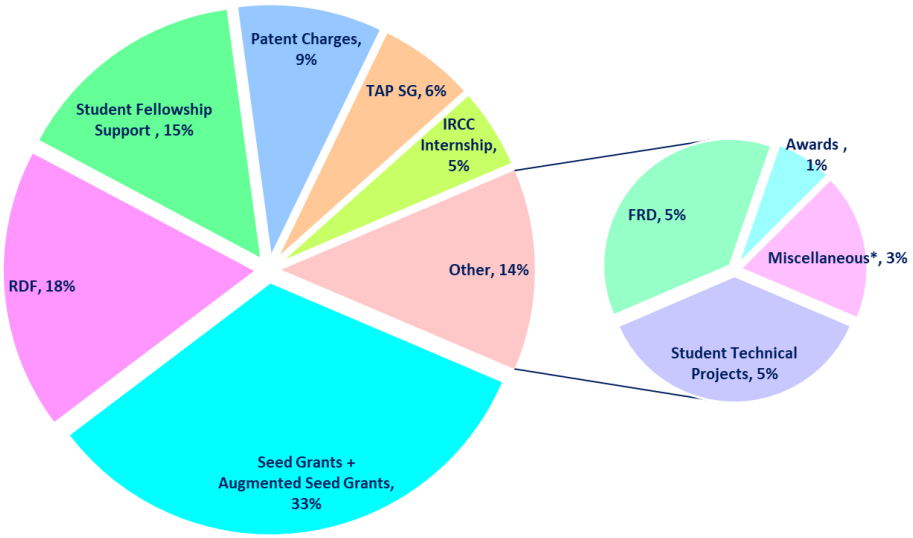
B. Internal Funding for R&D

The Institute released funds to the tune of Rs. 45 Crores for R&D related activities. The Institute provided internal funding for supporting faculty research. Of these funds, 15 Crore rupees were utilized for seed grant and augmented seed grant for initiation of research for new faculty members. This proactive approach ensures long-term institutional growth, jump-starts novel research ideas early, and rapidly expands our primary patent and publication pipeline. Allocating a dedicated 33% of internal funding toward early-career seed grants reveals an excellent strategy for nurturing young talent.

Overall, the research funds were distributed across the following areas:

- Seed grant for initiation of research for new faculty
- TAP (Teaching Assistantship – projects) Seed Grant
- Patent filing and renewal charges
- Awards and recognition
- Facilitating grants to faculty members for filing patents, research publications and/or international travel
- Student fellowship support
- Researcher Development Fund

- IRCC Research Internship
- Student Technical Projects



**FRD: Financial Support for R&D; RDF: Research Development Fund; TAP: Teaching Assistantship – Projects*

Percentage distribution of Institute funding related to R&D activities in FY 25-26

Ethics: IIT Bombay lays great emphasis on ethics as a part of its R&D efforts. The strict ethical compliance minimizes project risks and keeps the multi-disciplinary medical and biotech research well-aligned with global compliance standards. During 2025- 26, the Institute Ethics Committee received 32 new proposals and 27 proposals were approved in expedited, full board or exempt mode depending on the nature of the proposals. The Institutional Review Board (IRB) received 88 new proposals from various academic units and 92 proposals were approved and one was recommended to the Principal Investigator (PI) for submission to IEC. The Institute Animal Ethics Committee received 15 proposals, all of which were approved. The efficient review workflows demonstrate strong governance and a commitment to high international research standards.

Research Internship Awards: IRCC Research Internship is awarded to bright students across the country every year to undertake research at IIT Bombay for six months, based on national advertisement and selection process. By welcoming top external talent, the institute enriches its own research ecosystem while helping train the country's broader scientific workforce.

Each selected intern gets monthly stipend of Rs. 15,000 for six months (subject to them not getting any support from parent organization/government (MHRD)) along with contingency support of Rs. 75,000 per intern. In 2025, 3708 applications were received out of which 168

interns joined the program. A selective 4.5% acceptance rate highlights the exceptional quality and nationwide prestige of this program.

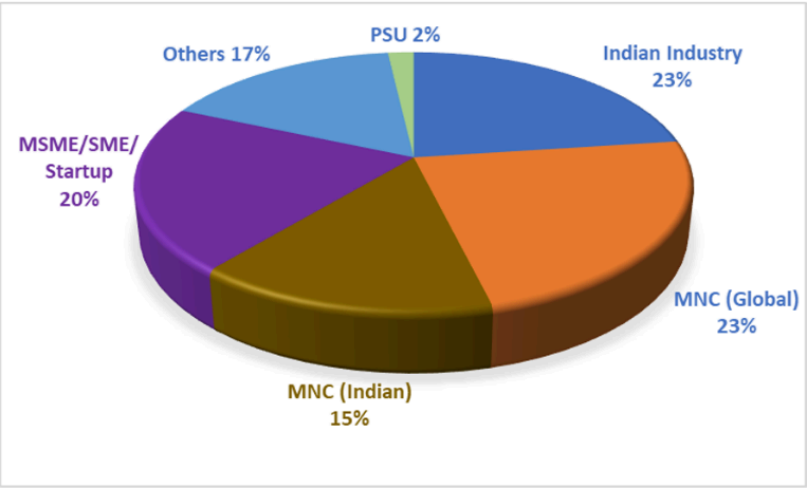
Online processes

The R&D management portal Drona was further updated this year incorporating new features, to improve IRCC services to IITB researchers. New modules for Institute Ethics Committee (IEC), Institutional Review Board (IRB), and Institute Animal Ethics Committee (IAEC) were released. Automation of IP Checklist for licensing, Annual Performance Assessment Report (APAR) module for assessment of IRCC staff, Seed Grant Review module, Central Facility usage model were also developed. The existing modules were also enhanced to make them more efficient. DRONA DTS (Document Tracking System) was extended to the Main Accounts section of IITB and integrated with the Cash Section for easier access to details, forwarding to the Cash Section, and tracking of records.

C. IIT Bombay – Industry Interactions for R&D

The Institute continued its efforts to promote interaction with industry collaborators.

Around 220+ Indian and international industries approached IIT Bombay for partnership through collaborative R&D projects and consultancy services. This continuous engagement deepens relationships with global industry leaders, creating collaborative projects that move academic insights directly into commercial applications. The interactions were managed seamlessly by facilitating exploratory meetings and technical discussions with faculty members in areas of mutual interest. Around 150 meetings, in both physical and virtual modes, were coordinated by IRCC during the year 2025-26. Agreements were entered into to establish a formal association. Managing over 150 targeted industry meetings reflects a highly professional corporate outreach operation.



Break-up of requests from different types of industries in FY 2025-26

Some of the notable industries/PSUs that led to projects being taken up include Asian Paints, Mercedes-Benz R&D India, Schlumberger, Dolby Laboratories, etc.



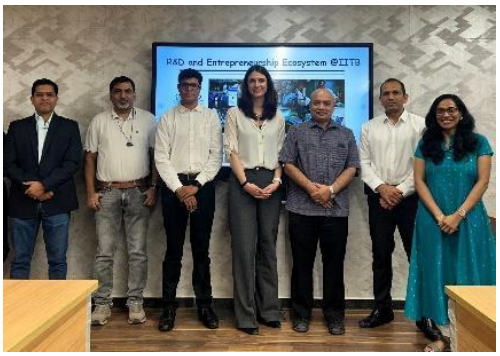
Meeting with Tata Power



Meeting with KPIT Technologies



Meeting with John Cockerill Group



Meeting with SLB Schlumberger



Meeting with Mitsui Chemicals

D. IRCC Stakeholder Outreach

SynergyX: IIT Bombay – SAMEER Workshop

IRCC organized a full-day SynergyX Workshop on January 8, 2026, at the SAMEER campus, IIT Bombay, to foster collaboration and explore joint R&D opportunities in advanced technologies. The partnership aimed to strengthen joint research, academic collaboration, and translational outcomes across areas such as medical devices, RADAR systems,

antennas, 5G/6G technologies, opto-electronics, quantum technologies, and environmental monitoring. It is expected to enable impactful research, shared infrastructure, active student engagement, and the development of translational research hubs. The workshop was led by Prof. Shireesh Kedare (Director, IIT Bombay) and Dr. P. Hanumantha Rao (Director General, SAMEER). MoU was also signed to formalize the association.




IITB- SAMEER Workshop: Lab visit and MoU signing during SynergyX

Visit by Mr. Rajesh Agarwal, Hon'ble Chief Secretary, Govt. of Maharashtra

IRCC hosted Shri. Rajesh Agarwal, Hon'ble Chief Secretary, Government of Maharashtra, on January 19, 2026, along with a senior delegation from the National Informatics Centre (NIC), and MahaIT, Government of Maharashtra. IIT Bombay's senior leadership led by Prof. Shireesh Kedare, Director, interacted with the delegates on ways to strengthen collaboration in education, startups, translational research, and innovation; fostering academia-industry partnerships; and exploring the deployment of IIT Bombay's sovereign LLM, BharatGen, across key sectors to drive India-centric innovation and societal impact. The Chief Secretary toured key facilities, including the Center for Semiconductor Technologies (SEMIX), Society for Innovation and Entrepreneurship (SINE), and the IIT Bombay Research Park.




Facility tour and discussions during Shri. Rajesh Agarwal's visit

Visit by Mr. B. Venugopal Reddy, Additional Chief Secretary, Higher and Technical Education Department, Government of Maharashtra

IRCC hosted Shri. B. Venugopal Reddy, Additional Chief Secretary and team, Higher and Technical Education Department (HTED), Government of Maharashtra on August 8, 2025.The delegation was received by Prof. Shireesh Kedare, Director IIT Bombay. During the visit, the delegation visited the IIT Bombay Nanofabrication Facility, the Drone Laboratory under the Maharashtra Drone Mission, the Makerspace (Deepak and Maya Satwalekar Lab), Society for Innovation and Entrepreneurship (SINE), the Technology Innovation Hub

(TIH), and the IIT Bombay Research Park on strengthening collaboration in higher education, startups, translational research, and innovation, as well as fostering academia–industry partnerships.



Shri. B. Venugopal Reddy interacting with IITB faculty members and researchers during his visit

E. Intellectual Property (IP) Protection Activities

The institute has filed 305 Intellectual property applications (including 264 patent applications) in FY 2025-26.

FY 25-26	YTD Total	
Type of IP	**Filed**	**Granted**
Indian Patent	249	147
Patent Cooperation Treaty (PCT)	5	NA
Foreign Patents	10	5
Trademark	9	12
Industrial Design	22	15
Copyright	10	6
Total IPs	305	185

IRCC conducted various awareness sessions for IP education. This supportive ecosystem gives researchers the clear, practical guidance needed to turn original concepts into legally protected, commercial-ready assets.

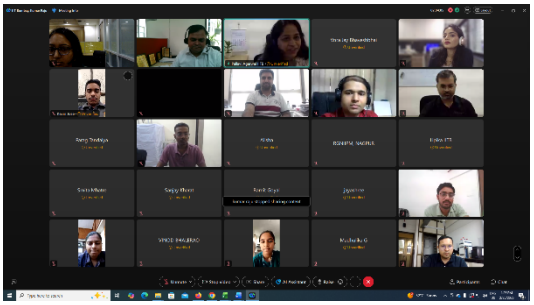
World Intellectual Property Day 2025 - World Intellectual Property Day was observed on Saturday, 26 April, 2025 at IIT Bombay. Prof. Shishir Kumar Jha gave a talk on “Copyright and the Emerging Digital Economy,” aligned with the WIPO theme “IP and Music: Feel the Beat of IP,”. Prof. Karuna Jain spoke on the “Importance of Intellectual Property in Academic

Institutions". The session concluded with an engaging question-and-answer session and a vote of thanks delivered by Prof. Patwardhan.

Webinar on Intellectual Property Rights (IPR) & Patent/Design Filing - On 25 February 2026, IIT Bombay, in collaboration with the Rajiv Gandhi National Institute of Intellectual Property Management (RGNIIPM), organized a webinar on Intellectual Property Rights (IPR) and Patent/Design Filing for students, staff, and faculty members. The objective of the webinar was to create awareness and provide practical guidance on intellectual property protection and filing procedures.

Session on Computer Related Inventions (CRI) and Intellectual Property Rights (IPR) Awareness - On 17 March 2026, an online session on Computer Related Inventions (CRI) and Intellectual Property Rights (IPR) awareness was jointly organized by IIT Bombay and S. Majumdar & Co. (SMCO). The session aimed to enhance awareness of intellectual property rights and computer-related inventions among participants.



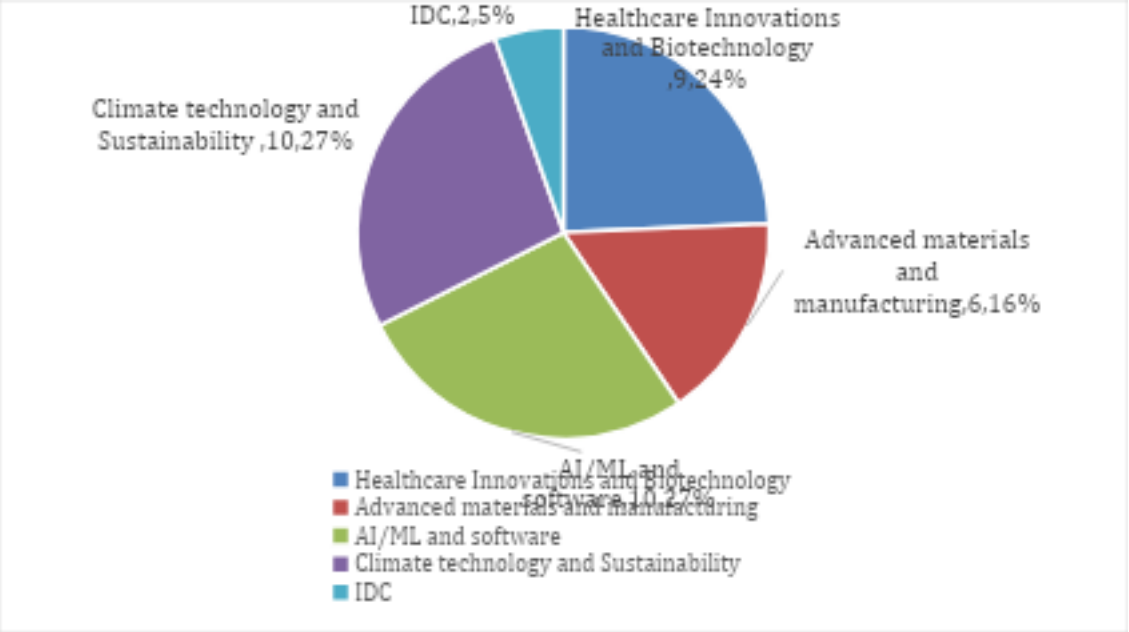


IP related outreach activities in FY 25-26

F. Technology Licensing Activities in FY 2025-26

In FY 2025–26, IIT Bombay licensed 37 technologies to various industries and startups. The total value of the licensing deals signed was ₹16.78 crore. Securing ₹16.78 crore in licensing deals shows excellent market validation of the institute's technology. These projects deliver clear social and economic impact by putting practical solutions—like agricultural coolers and clean-tech carbon capture tools—directly into public and industrial use.

Domain wise break up of licensed technologies is given below:



Domain wise break up of licensed technologies in FY 25-26

Glimpses of Technologies licensed and commercialized

A system to remove floating debris in open water channels

IIT Bombay researchers have developed a system for removing floating debris in an open water channel. Designed to minimize manual intervention, the technology effectively eliminates the need for manual garbage removal and enhances operational efficiency. The patented system has been successfully demonstrated and installed on-site in collaboration with industry partner. IIT Bombay has licensed this technology to industry for commercialization and large-scale deployment.

Multi-Utility Heat Pump Dryer and Cooler

IIT Bombay researchers have developed a technology related to “Multi-Utility Heat Pump Dryer and Cooler” with a nominal cooling capacity of 0.5 to 1.5 TR. for rural applications. The system enables simultaneous cooling and drying of agricultural produce thereby extending shelf life, minimizing post-harvest losses, and enabling value addition. The system is energy-efficient, environmentally friendly, and economically viable. The technology has been licensed to industry for market deployment.

inSPEC (indigenous pulsed eddy current) instrument

IIT Bombay researchers have developed a technology related to “inSPEC (indigenous pulsed eddy current) instrument,” for non-invasive inspection of steel rebars in concrete structures using pulsed eddy current. This hand-held device is used for non-destructive testing and inspection. The inventors have founded the startup for further product development and commercialization.

Docked digital platform

IITB Researchers have developed a mobile application called "*Docked*" for medical professionals to digitally record and manage logbook entries. The platform offers specialty-specific digital logbooks, real-time dashboards, automated report generation, and integrated faculty approval systems, thereby transforming documentation and workflow management in medical education and healthcare practice. The technology is being commercialized by the inventors through their startup for market deployment.

Combining direct air capture and soda ash plant processing for sustainable CO2 mitigation

IIT Bombay researchers have developed a technology related to "Combining direct air capture and soda ash plant processing for sustainable CO2 mitigation". The technology relates to the direct air capturing technique to capture carbon dioxide from the atmosphere. The technology introduces a novel zinc-metal catalyst that provides a robust, selective, and efficient solution for CO2 capture. To further develop and commercialize the technology, the inventors have established a clean-tech startup focused on sustainable carbon mitigation solutions and industrial deployment.

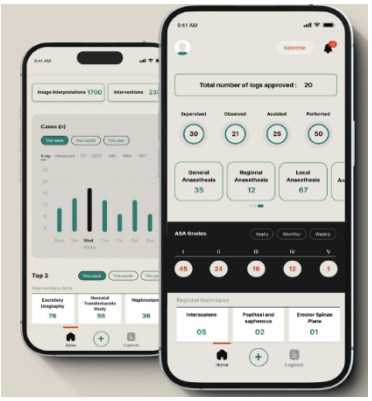




License agreement signing with Ainwesha for "removal floating debris in open water channels" Multi-Utility Heat Pump Dryer and Cooler



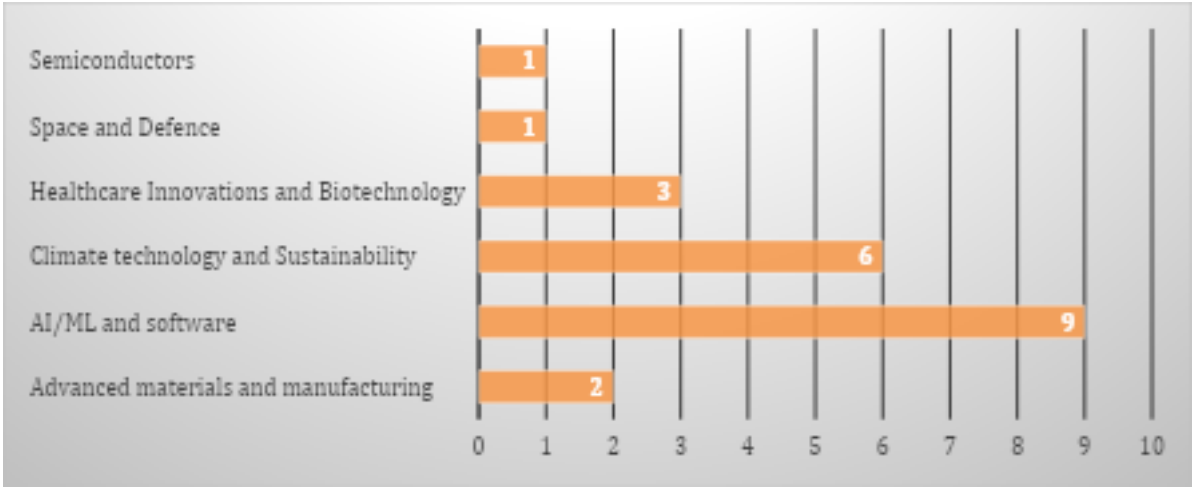
inSPEC instrument prototype



Docked platform user interface

G. Support to Entrepreneurship

The close productive framework between IITB and SINE empowers researchers to scale up their innovations, turning academic intellectual property into growing companies that drive job creation and market innovation. In FY 2025–26, IIT Bombay granted 40 in-principle approvals/No Objection Certificates (NoCs) to researchers for the formation of companies and new ventures, and for the use and commercialization of technologies and intellectual property developed at the Institute. Among these ventures, 22 startups formally incubated at Society for Innovation and Entrepreneurship (SINE), highlighting the vibrant entrepreneurial ecosystem and strong innovation culture at the Institute.



Domain wise break up of start-ups in FY 25-26

H. R&D Related Agreements

During the financial year 2025–26, more than 380 agreements and Memorandum of Understanding (MoUs) were successfully finalized and executed. These included a wide range of engagements such as research collaboration agreements, non-disclosure agreements (NDAs), Student Sponsorship Agreements, Service Agreements, Consultancy Agreements, Grant Agreements, and Material Transfer Agreements (MTAs).

These collaborations were established with a diverse set of stakeholders, including leading industries, reputed organizations, academic institutions, and government bodies, both at the national and international levels. The volume of such research and collaborative activities has demonstrated strong growth, with a year-on-year increase of 18%. The breadth and diversity of these agreements underscore a strong institutional commitment to fostering strategic collaborations, advancing research and innovation, and strengthening engagement between academia, industry, and government. An 18% growth in finalized agreements reflects highly efficient administrative and legal operations. This agility reduces bureaucratic friction, allowing collaborative research with global partners to get underway quickly and smoothly.

The list major MOU's/Agreements finalised during this financial year are Infosys Limited, Engineering Projects India Ltd., Tech Mahindra, Central Railway, ROHM Semiconductor India Private Limited amongst others.



Maharashtra Maritime Board, Government of Maharashtra



Rolls-Royce India Private Limited



BP International Limited





Maharashtra State Electricity Transmission Company Ltd. AIIFA Sustainable Steel Manufacturers Association

I. Awards and Recognition

Dr. P. K. Patwardhan Technology Development Award

For the year 2024, the Dr. P. K. Patwardhan Technology Development Award was conferred on Prof. Rajesh Zele and Dr. Vijay Kanchetla from the Department of Electrical Engineering for their work DhruvaPro. The award was presented on 15th September 2025 at IIT Bombay's - National Engineer's Day celebrations.

Prof. Zele and Dr. Kanchetla was felicitated for their exemplary contribution DhruvaPro – A Worldwide Navigational Receiver Chip for NAVIC, GPS, GALILEO & BEIDOU. DhruvaPro is the first indigenous silicon-proven RF receiver chip compatible with NavIC, GPS, Galileo, and BeiDou. Developed at IIT Bombay, it enables precise global navigation by capturing weak satellite signals with high sensitivity. The innovation has been successfully transferred to MumbaiSemi, an IIT Bombay startup, for commercialization. The awardees presented their work to the Institute community by delivering a talk organized by IRCC in October 2025.

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Prof. Rajesh Zele and Dr. Vijay Kanchetla receiving Dr. P. K. Patwardhan Technology Development Award; Prof. Sujoy Bhore recipient of Prof. Krithi Ramamritham Award for Creative Research, delivering a talk

Prof. Krithi Ramamritham Award for Creative Research

The Prof. Krithi Ramamritham Award for Creative Research for 2024 was conferred on Prof. Sujoy Bhore, Department of Computer Science and Engineering, for his creative work on "Theoretical Computer Science" on National Education Day celebrated by the institute on 11th November 2025. This award consisted of a citation and a cash prize of Rs. 1,00,000/- given to the awardee. The awardee presented his work to the Institute community by delivering a talk organized by IRCC in December 2025.

IRCC Research Awards

The IRCC Awards are instituted and funded with the objective of recognizing excellence in research and consultancy work that has made outstanding contributions and demonstrated significant impact. The awards are presented across four categories.,

The awardees of Research Publication Award 2024 are

1. Prof. Hrishikesh Gadgil, Dept. of Aerospace Engineering
2. Prof. Santanu K. Ghosh, Dept. of Biosciences and Bioengineering
3. Prof. Rajarshi Chakrabarti, Dept. of Chemistry
4. Prof. Swagata Basu, Dept. of Civil Engineering
5. Prof. Vikram Vishal, Dept. of Earth Sciences

The awardees of Research Dissemination Award 2024 are

1. Prof. Pradip Kalbar, Dept. Environmental Science and Engineering
2. Prof. Swatantra Pratap Singh, Dept. Environmental Science and Engineering

The awardee of Impactful Research Award 2024 is

1. Prof. Manish Kumar, Dept. of Civil Engineering

The awardees of Early Research Achiever Award 2024 are

1. Prof. Srinivasan Ramakrishnan, Dept. of Chemistry
2. Prof. Swathy Manohar, Dept. of Civil Engineering
3. Prof. Gurminder Singh, Dept. of Mechanical Engineering



Recipients of IRCC Research Awards with Director, IIT Bombay and other dignitaries on National Education Day

These awards were conferred to the awardees on National Education Day celebrated by the institute on 11th November 2025. Each award consisted of a citation and a cash prize of Rs. 50,000/- given to the awardee. In addition, the awardee(s) will be invited to submit a research proposal to IRCC for possible funding of up to Rs. 5,00,000/-. The awardees presented their work to the Institute community by delivering talks organized by IRCC in November and December 2025.

Awards for Excellence in Research

Prof. S. C. Bhattacharya Award for Excellence in Pure Sciences and Prof. H. H. Mathur Award for Excellence in Applied Sciences are one-time recognitions given to faculty members for the research they have carried out at IIT Bombay over the past ten years.

The recipients for the year 2025 are as follows:

- Prof. K. G. Suresh, Department of Physics - Prof. S.C. Bhattacharya Award for Excellence in Pure Sciences 2025
  Title: Magnetism in Intermetallics, Spintronics, Magnetic Skyrmions, Topological Materials
- Prof. Radhey Shyam Jangid, Department of Civil Engineering - Prof. H. H. Mathur Award for Excellence in Applied Sciences 2025
  Title: Seismic Base Isolation, Earthquake-Resistant Design, Stochastic Earthquake Analysis, Inerter-Based Control Devices

These awards were conferred to the awardees on the Institute Foundation Day celebrated on 10th March 2026. The awardees presented their work to the Institute community by delivering talks organized by IRCC in March 2026.



Prof. K. G. Suresh receiving Prof. S.C. Bhattacharya Award for Excellence in Pure Sciences 2025; Prof. R. S. Jangid receiving Prof. H. H. Mathur Award for Excellence in Applied Sciences 2025

External R&D and IP Awards

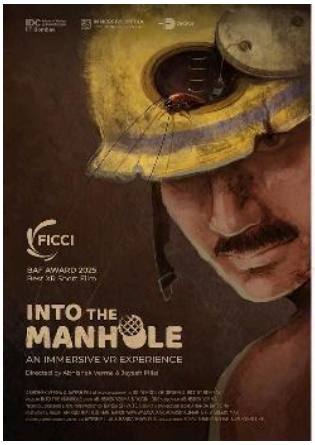



India's Breakthrough CAR-T Therapy recognized among Top 3 National Innovations





IIT Bombay receives ICMR Innovation Excellence Recognition – Gold for Surgical Robotics Patent; IIT Bombay receives Questel IP Excellence Award 2025







Into The Manhole', directed by Prof. Abhishek Verma and Prof. Jayesh Pillai, wins FICCI BAF Award 2025 for Best XR Short Film; Mr. Dhimant Vyas, Professor of Practice,IDC wins India's Best Design Award 2025 for Title animation of feature film 'Sitaare Zameen Par'

J. R&D Events

IRCC carries out various large-scale events bridge the gap between academic progress and the needs of industry and national policy. High-profile interaction with various industries and policy leaders etc. helps align our research direction with major national missions like Atmanirbharta (self-reliance) and clean energy development.

TechConnect 2025

TechConnect 2025, a flagship event showcasing IIT Bombay's R&D activities, was organised by the Office of the Dean (R&D) from 22nd to 24th December 2025. This event aimed at showcasing technological and groundbreaking R&D achievements of IIT Bombay through Exhibition, Department-Industry Symposia and ResCon, the key verticals of the event.

The Exhibition showcased a diverse range of exhibits from multiple academic units, featuring 140 working models, demonstrations, and products that highlighted our technological innovations. The Industry–Department Symposia comprised 11 focused sessions, providing a platform for close interaction, in-depth discussions, and firsthand exposure to departmental technologies and research facilities. ResCon, a research conclave, showcased student-driven projects and hosted the Soch Vichar Achar competition, giving undergraduates across India a platform to present innovative research.







Glimpses from TechConnect 2025

Mr. N. R. Narayana Murthy, Founder of Infosys Ltd. and Catamaran Ventures, inaugurated TechConnect 2025, and visited exhibits showcasing technologies in manufacturing, engineering, biosciences, game design, defence, artificial intelligence, and more.

The Chief of Defence Staff, General Anil Chauhan, visited TechConnect 2025, engaging with innovators and exploring technologies across various domains, accompanied by Lt. General D. S. Kushwaha, GoC Western Region. They toured defence-focused stalls, promoting self-reliance ('atmanirbharta') and supporting the future readiness of India's armed forces. Overall, TechConnect attracted an overwhelming footfall, with visitors displaying keen enthusiasm and excitement for innovations in science and technology.

Industry Day - Intelligent Decision Support Systems (IDSS)

On 10th May 2025, IRCC, in collaboration with the Technocraft Center for Applied Artificial Intelligence (TCA2I), hosted the Industry Day on Intelligent Decision Support Systems (IDSS). The event centered on pioneering research in Defence Innovation, Medical Informatics, and Smart XR technologies. The event saw strong participation from industry stakeholders, contributing to its overall success and reinforcing the value of collaborative innovation in the field of intelligent decision support systems.



Lab visit and Industry Talk during IDSS

Industry Day - Indoor Air Quality Management (IAQM)

IIT Bombay's Industrial Research and Consultancy Centre (IRCC) successfully hosted the Industry Day on Indoor Air Quality Management (IAQM) on 28th June 2025. The day featured in-depth discussions on cutting-edge research in areas such as IoT-based sensor networks, computational fluid dynamics (CFD), regulatory frameworks governing indoor air quality (IAQ), and the development of full-scale testbeds for real-world IAQ experimentation. The event was well-received, with active involvement from representatives across the relevant industries contributing to its success.



Lab visit and panel discussion during IAQM

Industry Day - 'Empowering India's Solar Future (EISF)

IRCC and National Centre for Photovoltaic Research and Education (NCPRE) hosted the Industry day on 'Empowering India's Solar Future (EISF) - Innovation, Indigenisation, and Collaboration' on 8th November 2025. Shri Santosh Kumar Sarangi, Secretary, Ministry of New and Renewable Energy (MNRE), graced the event as the Chief Guest. Shri Sarangi inaugurated the ART-PV India and Module Simulator Facility at NCPRE, further enhancing India's capabilities in renewable energy research, development, and testing infrastructure.

The event brought together leaders from academia, industry, and policy to discuss key areas such as Solar Deployment and Manufacturing, Next-Gen PV, and Solar Grid Integration. It featured insightful talks highlighting IIT Bombay's research, followed by engaging panel discussions with industry experts on shaping future pathways in these domains. Visits to IIT Bombay's Silicon Fabrication Lab, Characterization Lab, Tandem Device Lab, Module Lab, and Power Electronics Lab, were organized.



Lab visit and panel discussion during EISF

International Conference on New and Emerging Concepts (ICONS)

IRCC held its flagship workshop - International Conference on New and Emerging Concepts (ICONS) on 20th - 21st Feb, 2026. The theme chosen for this year, Critical Raw Materials.

IIT Bombay has been recognised as a "A Centre of Excellence in Critical Minerals, Metals and Materials (CMMM)" under the National Critical Mineral Mission. In line with this National Mission, the workshop leveraged and showcased the unique strength of IIT Bombay, spanning across the entire value chain of Mines-Minerals-Materials-Magnets-Market. Attended by over 175 participants, the workshop catalysis extensive discussions and opened new collaborative opportunities with a wide spectrum of stakeholders from academia, government, industry and policymakers.

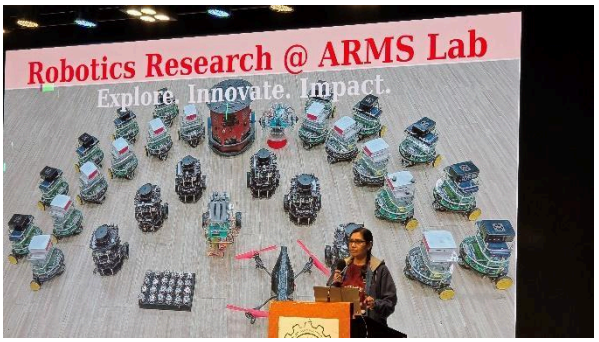


Delegates at ICONS 2025

Enthuse 2025, The Freshman Research Orientation at IIT Bombay

Enthuse 2025, held on 24th July 2025 as part of the Undergraduate Orientation Programme at IIT Bombay, was a vibrant introduction to the institute's research landscape for the incoming undergraduate batch. The event, organized by IRCC, took place simultaneously across three venues on the IIT Bombay campus and was led by three faculty coordinators: Prof. Abhijit Majumder (Dept. of Chemical Engg.), Prof. Amber Shrivastava (Dept. of Mechanical Engg.) and Prof. Arpita Mondal (Dept. of Civil Engg.). During the event, professors from diverse departments presented brief insights into their research, offering a snapshot of the broad spectrum of research activities at IIT Bombay. Overall, Enthuse 2025 provided an enriching platform for students to engage with faculty, fostering enthusiasm for research from the very beginning of their academic journey.





Enthuse Research Talks

K. Dissemination and Outreach for R&D

Lectures on Nobel Prizes 2025

As part of the 2025 lecture series titled as “aisa kya kiya in logo ne?”, faculty members delivered talks on the Nobel Prizes across various disciplines. Prof. Kantimay Das Gupta delivered an in-depth lecture on the Nobel Prize in Physics titled *“SLUGS, SQUIDS and Macroscopic Variables in Quantum Mechanics,”*. Prof. Sushil Kumar presented a talk on the Nobel Prize in Physiology or Medicine titled *“Discovery of Central Immune Tolerance Mechanism that Prevents Harming the Body,”* . The Nobel Prize in Chemistry lecture was delivered by Prof. Sanjog Nagarkar, in his talk titled *“Metal Organic Frameworks: Why the World’s Smallest Lego Set Matters,”*. The Nobel Prize in Economic Sciences was covered by Prof. Pintu Parui through his lecture titled *“Innovation-Driven Economic Growth: Why It Matters,”*

Research News Articles total no of press counts

IRCC realizes the need to communicate its R&D efforts with the wider society. Towards this, IRCC brought out 42 easily accessible stories this financial year, conveying IIT Bombay’s achievements. These articles have been released in English, Hindi and Marathi languages. These articles were reported in 231 media releases across three languages, boosting the visibility of IIT Bombay’s vibrant R&D ecosystem.


ET Healthworld.com

IIT Bombay develops DNA-based strategy to tackle antibiotic resistance

In the first study, the researchers focused on specially identified short strands of DNA known as aptamers. Unlike the conventional drugs, aptamers are made up of nucleic acids, synthetically produced, relatively stable, and easier to modify.

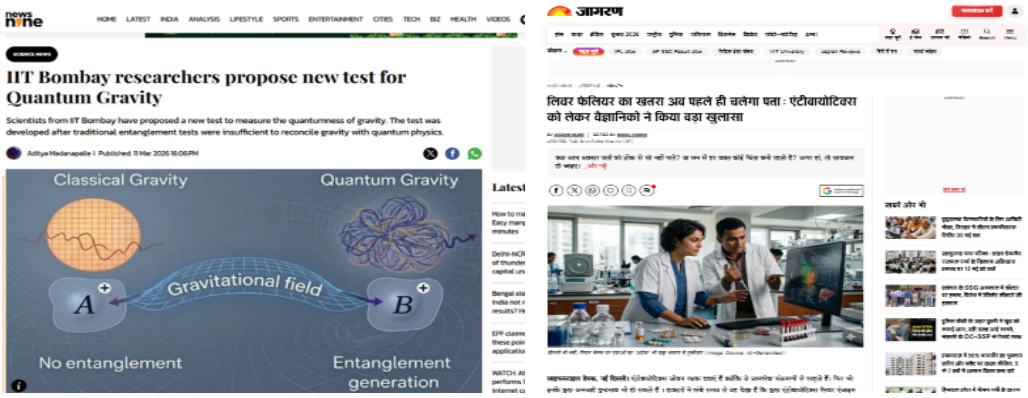
Mumbai: Researchers at the Indian Institute of Technology Bombay (IIT Bombay) have developed a DNA-based strategy that can make drug-resistant


सकाळ

New TB Medicine Breakthrough: IIT मुंबईचा शोध ठरणार गेमचेंजर! आता TBची औषधं देणार जलद परिणाम

क्षयरोगाची औषधे होणार प्रभावी

IIT Bombay TB research



R&D Publications

The brochure titled 'R&D Highlights', reporting some of the outcomes of R&D efforts undertaken at the Institute, interactions with the industry, and the possible modes of collaboration, was updated. The IRCC newsletter titled 'Research & Innovations in Sciences and Engineering (RISE)' was brought out to showcase creative, innovative, and high-impact research emerging from the Institute's ecosystem across focused thematic areas, like Semiconductors and Robotics and AI. These publications were disseminated to industry partners to enhance awareness of the Institute's R&D capabilities and to strengthen engagement with external stakeholders.



L. Student Technical Projects

This section lists outstanding global rankings achieved by our student teams

Autonomous Aerial Systems & Drone Teams

Team AeRoVe secured 7th rank at ICUAS 2026, completed all objectives in the Technoxian Drone Rescue Challenge 2025, and qualified for the UAS Challenge 2026 hardware round. Team ExoFly secured 23rd place in Technical Design at SUAS 2025 and achieved 100% survivor identification at NIDAR 2026. Team Rakshak won the Design Report Award at CUASC 2025 and developed a dual-drone search-and-rescue solution at NIDAR 2025.

Team AeRoVe- From grass fields to global arenas; Team ExoFly at NIDAR UAS; Team Rakshak at CUASC 2025, California, USA with the Nimbus drone

Autonomous Mobility & Racing Teams

Team SeDriCa secured 3rd place at RoboRacer and developed a full-scale Autonomous Buggy. IIT Bombay Racing secured the Overall 2nd Runner-Up position at Formula Bharat 2026 and showcased India's first fully autonomous Formula Student car. IIT Bombay Racing Driverless secured 4th place overall at FSAI 2025 and became the first Indian team to complete a dynamic event.

SeDrica's autonomous buggy inaugurated by Shri Nitin Gadkari Ji; IITB Racing-Formula Student AI 2025; IITB Racing Driverless- Overall 4th -Formula Student AI 2025

Aerospace, Robotics & Space Exploration Teams

Team Rocket successfully completed a launch at IREC 2025 and was recognized for its avionics design, AUV-IITB secured top positions at RoboSub 2025, and IITB MARS Rover achieved 2nd position globally at ERC 2025 – Remote Edition.

Team Rocket-Team photo after successful flight safety review; Team AUV-IITB at RoboSub'25; IITB Mars- IRC 2026

Sustainability, Chemical Engineering & Synthetic Biology Teams

Team ChemEca ranked among the top 7 global teams in the Open Air Carbon Removal Challenge 2026, Team iGEM won Silver Medals at international competitions, and Team SHUNYA won Solar Decathlon 2024 and became a Top-6 finalist at Solar Decathlon India 2025–26.

Team ChemEca at Global Sustainability Challenge (GSC) Exhibition; Team iGEM IIT Bombay 2025 bags silver medal at the Grand Jamboree in Paris; Team SHUNYA- Launch of 6th edition of team's Annual Newsletter during Team's Annual Net-Zero Technical Conclave, SUSTAIN 2026.

The outcomes documented in this report reflect the collective effort of our researchers, students, and collaborators — each result a stepping stone opening new scientific questions and engineering possibilities. We remain committed to curiosity-driven inquiry alongside application-focused development, deepening our investigative threads and expanding collaborations with academia, industry, and government. Most importantly, we look forward to nurturing the next generation of researchers who will carry this work into exciting new frontiers.